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*                                                  *
*           STAAD.Pro                              *
*           Version  2006     Bld 1005.US          *
*           Proprietary Program of                 *
*           Research Engineers,  Intl.             *
*           Date=    MAY 28, 2007                  *
*           Time=    11:28:17                      *
*                                                  *
*      USER ID: MIPL                               *
****************************************************

     1. STAAD PLANE
INPUT FILE: External Staircase.STD
     2. START JOB INFORMATION
     3. JOB NAME - DUBAI METRO PROJECT
     4. JOB CLIENT - GOVERNMENT OF DUBAI - ROADS _TRANSPORT AUTHORITY
     5. ENGINEER NAME CJA
     6. ENGINEER DATE 26-MAY-07
     7. JOB NO 068010.21
     8. END JOB INFORMATION
     9. INPUT WIDTH 79
    10. UNIT METER KN
    11. JOINT COORDINATES
    12. 1 0 0 0; 2 -3.36 2.3 0; 3 -4.56 2.3 0; 4 0.56 5 0; 5 2.56 5 0
    13. MEMBER INCIDENCES
    14. 1 1 2; 2 2 3; 3 2 4; 4 4 5
    15. DEFINE MATERIAL START
    16. ISOTROPIC STEEL
    17. E 2.05E+008
    18. POISSON 0.3
    19. DENSITY 76.8195
    20. ALPHA 1.2E-005
    21. DAMP 0.03
    22. END DEFINE MATERIAL
    23. MEMBER PROPERTY BRITISH
    24. 1 TO 4 TABLE ST CH260X75X28
    25. CONSTANTS
    26. MATERIAL STEEL ALL
    27. SUPPORTS
    28. 1 2 4 5 PINNED
    29. LOAD 1 SELFWEIGHT
    30. SELFWEIGHT Y -1
    31. LOAD 2 SDL
    32. MEMBER LOAD
    33. 1 TO 4 UNI GY -1.
    34. LOAD 3 LL
    35. MEMBER LOAD
    36. 1 TO 4 UNI GY -2.5
    37. LOAD COMB 1001 1.4DL + 1.6LL
    38. 1 1.4 2 1.4 3 1.6
    39. LOAD COMB 2001 1.0DL - 1.0LL
    40. 1 1.0 2 1.0 3 1.0
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    41. LOAD COMB 2002 1.0DL + 0.5LL
    42. 1 1.0 2 1.0 3 0.5
    43. PERFORM ANALYSIS

              P R O B L E M   S T A T I S T I C S
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     NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS =      5/      4/      4
     ORIGINAL/FINAL BAND-WIDTH=       2/      2/      5 DOF
     TOTAL PRIMARY LOAD CASES =      3, TOTAL DEGREES OF FREEDOM =       7
     SIZE OF STIFFNESS MATRIX =         1 DOUBLE  KILO-WORDS
     REQRD/AVAIL. DISK SPACE  =      12.0/ 395121.6 MB

    44. PARAMETER 1
    45. CODE BS5950
    46. TRACK 1 ALL
    47. PY 375000 ALL
    48. SBLT 0 ALL
    49. CHECK CODE ALL

                        STAAD.Pro CODE CHECKING - (BSI )
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   PROGRAM CODE REVISION V2.9_5950-1_2000
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ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY             MZ       LOCATION
=======================================================================

      1 ST CH260X75X28    PASS     BS-4.8.3.3.1       0.270      1001
                          5.54 C         0.00          8.86        -
|-----------------------------------------------------------------------|
| CALCULATED CAPACITIES FOR MEMB     1 UNIT - kN,m  SECTION CLASS  1   |
|MCZ=  123.0 MCY=   19.3 PC=   191.4 PT=  1316.2 MB=   37.7 PV=  409.5|
| BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 1.00, my = 1.00, myx = 0.00 |
| PZ=  1316.25    FX/PZ =   0.00      MRZ=  123.0       MRY=   23.2   |
|-----------------------------------------------------------------------|
      2 ST CH260X75X28    PASS     BS-4.3.6           0.041      1001
                          0.00           0.00          4.16        0.00
|-----------------------------------------------------------------------|
| CALCULATED CAPACITIES FOR MEMB     2 UNIT - kN,m  SECTION CLASS  1   |
|MCZ=  123.0 MCY=   19.3 PC=   851.4 PT=     0.0 MB=  101.0 PV=  409.5|
| BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 1.00, my = 1.00, myx = 0.00 |
| PZ=  5854.97    FX/PZ =   0.00      MRZ=  123.0       MRY=   23.2   |
|-----------------------------------------------------------------------|
      3 ST CH260X75X28    PASS     BS-4.8.3.3.1       0.344      1001
                          7.80 C         0.00          9.46        -
|-----------------------------------------------------------------------|
| CALCULATED CAPACITIES FOR MEMB     3 UNIT - kN,m  SECTION CLASS  1   |
|MCZ=  123.0 MCY=   19.3 PC=   143.7 PT=  1316.2 MB=   32.6 PV=  409.5|
| BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 1.00, my = 1.00, myx = 0.00 |
| PZ=  1316.25    FX/PZ =   0.01      MRZ=  123.0       MRY=   23.2   |
|-----------------------------------------------------------------------|
      4 ST CH260X75X28    PASS     BS-4.3.6           0.132      1001
                          0.00           0.00          9.46        0.00
|-----------------------------------------------------------------------|
| CALCULATED CAPACITIES FOR MEMB     4 UNIT - kN,m  SECTION CLASS  1   |
|MCZ=  123.0 MCY=   19.3 PC=   639.4 PT=     0.0 MB=   71.9 PV=  409.5|
| BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 1.00, my = 1.00, myx = 0.00 |
| PZ=  5854.97    FX/PZ =   0.01      MRZ=  123.0       MRY=   23.2   |
|-----------------------------------------------------------------------|

  ************** END OF TABULATED RESULT OF DESIGN **************

   50. LOAD LIST ALL
   51. PRINT SUPPORT REACTION LIST 1 2 4 5
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SUPPORT REACTIONS -UNIT KN   METE    STRUCTURE TYPE = PLANE
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JOINT  LOAD   FORCE-X   FORCE-Y   FORCE-Z    MOM-X    MOM-Y    MOM Z

  1     1      -0.01      0.53      0.00      0.00     0.00     0.00
        2      -0.05      1.96      0.00      0.00     0.00     0.00
        3      -0.12      4.91      0.00      0.00     0.00     0.00
     1001      -0.28     11.35      0.00      0.00     0.00     0.00
     2001      -0.19      7.40      0.00      0.00     0.00     0.00
     2002      -0.12      4.95      0.00      0.00     0.00     0.00
  2     1       0.05      1.47      0.00      0.00     0.00     0.00
        2       0.20      5.47      0.00      0.00     0.00     0.00
        3       0.50     13.67      0.00      0.00     0.00     0.00
     1001       1.16     31.59      0.00      0.00     0.00     0.00
     2001       0.76     20.61      0.00      0.00     0.00     0.00
     2002       0.51     13.78      0.00      0.00     0.00     0.00
  4     1      -0.04      1.19      0.00      0.00     0.00     0.00
        2      -0.15      4.42      0.00      0.00     0.00     0.00
        3      -0.38     11.05      0.00      0.00     0.00     0.00
     1001      -0.87     25.53      0.00      0.00     0.00     0.00
     2001      -0.57     16.65      0.00      0.00     0.00     0.00
     2002      -0.38     11.13      0.00      0.00     0.00     0.00
  5     1       0.00      0.05      0.00      0.00     0.00     0.00
        2       0.00      0.18      0.00      0.00     0.00     0.00
        3       0.00      0.45      0.00      0.00     0.00     0.00
     1001       0.00      1.05      0.00      0.00     0.00     0.00
     2001       0.00      0.69      0.00      0.00     0.00     0.00
     2002       0.00      0.46      0.00      0.00     0.00     0.00

************** END OF LATEST ANALYSIS RESULT **************

   52. LOAD LIST ALL
   53. PRINT FORCE ENVELOPE NSECTION 5 ALL
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